



Opening the Black Box of "Not-Invented-Here": Attitudes, Decision Biases, and Behavioral Consequences

Journal:	<i>Academy of Management Perspectives</i>
Manuscript ID:	AMP-2013-0091-A.R3
Document Type:	Article
Keywords:	Organizational Learning < Organization and Management Theory < Topic Areas, Behaviors and Characteristics of Technical Professionals < Technology and Innovation Management < Topic Areas, Resistance to change < Organizational Development and Change < Topic Areas, Knowledge based management < Global Strategy and Tactics < International Management < Topic Areas, Technology and Innovation Management < Topic Areas, Decision making < Individual Level of Analysis < Organizational behavior < Topic Areas

**Opening the Black Box of "Not-Invented-Here":
Attitudes, Decision Biases, and Behavioral Consequences**

David Antons

RWTH Aachen University

antons@time.rwth-aachen.de

Frank T. Piller

RWTH Aachen University

piller@time.rwth-aachen.de

Abstract

The Not-Invented-Here (NIH) syndrome describes a negative attitude towards knowledge (ideas, technologies) derived from an external source. Despite being one of the most cited constructs in the literature on knowledge transfer, previous research has not provided a clear understanding of the antecedents, underlying attitudes, and behavioral consequences of NIH. The objective of our paper is to open the black box of NIH by providing an in-depth analysis of this frequently mentioned, yet rarely understood phenomenon. Building on recent research in psychology and an extensive review of the management literature on NIH, we first develop a framework of different sources classifying knowledge as "external" which might trigger a rejection of such inputs. Differentiating various functions of an attitude, we then identify possible trajectories linking NIH with individual behavior and decision-making. We apply this understanding to develop an extensive agenda for future research.

Keywords

Not-invented-here syndrome, open innovation, knowledge transfer, knowledge exchange, decision-making biases

Acknowledgements

We thank Ruth Jiang for her assistance in the literature coding process. Our colleague Iring Koch provided great advise on the psychological background of NIH. We also benefited from valuable comments by Torsten-Oliver Salge, Dirk Lüttgens, and Robin Kleer on earlier drafts of this manuscript. We acknowledge financial support for this study by DFG (German Research Foundation) within the "German Excellence Initiative".

Introduction

Staying innovative over time is a major challenge for organizations and individuals alike. Since March's (1991) seminal work, management literature has discussed the managerial trade-off of exploiting routines and core capabilities while simultaneously exploring innovative products and business opportunities. However, core capabilities are frequently a major source of rigidities when it comes to innovation and change (Benner & Tushman, 2003; Leonard-Barton, 1992). To become better at exploration, studies unanimously emphasize the need for an organization to successfully transfer and absorb outside knowledge as a potent driver of innovative output, firm performance, and economic welfare (e.g. Laursen & Salter, 2006; Lichtenthaler, 2011).

Previous research has demonstrated that this is not an easy task. Organizational inertia and structural rigidities challenge the transfer and utilization of outside knowledge on the level of the organization (Lane, Koka, & Pathak, 2006; Zahra & George, 2002). In most instances, however, knowledge is actually transferred, absorbed, and put into practice on an individual level (Lichtenthaler, 2011; Reagans & McEvily, 2003; Rogan & Mors, 2014). Here, previous research has identified multiple heuristic concepts influencing and biasing knowledge utilization and decision-making on the level of the individual, including representativeness, anchoring, and availability (Kahneman & Tversky, 1979), escalating commitment (Staw, 1981; Zardkoohi, 2004), or an endowment effect (Plott & Zeiler, 2005). Other literature, especially research in social psychology, has shown that in situations characterized by interactions and exchanges with external entities or external objects the attitudes of individuals often affect decision-making and lead to biased behavior (Ajzen, 2001; Bohnert & Wänke, 2002).

When it comes to absorbing external knowledge for innovation, the most frequently mentioned bias influencing individual decision-making is the **Not-Invented-Here Syndrome (NIH)**. It can

1
2
3 be best conceived of as a profound attitude-based bias towards knowledge (ideas, technologies)
4
5 derived from a source or contextual background that is considered "outside" or "external" from
6
7 the perspective of the individual (Katz & Allen, 1982; Kostova & Roth, 2002). There are ample
8
9 reasons why knowledge in this context is perceived as “external”, including a developer talking
10
11 to a team member who has a different disciplinary background, a colleague from a neighboring
12
13 department suggesting an idea, an external technology provider offering a technical solution, or a
14
15 customer from a different cultural tradition. Research on NIH postulates that individuals have a
16
17 generally negative attitude towards such knowledge, ideas, or technologies of external origin
18
19 (Burcharth, Knudsen, & Søndergaard, 2014; Laursen & Salter, 2006). When this predisposition
20
21 holds irrespective of the objective value of an external input, an individual is said to be affected
22
23 by the NIH syndrome. For an innovating organization, this bias becomes economically damaging
24
25 when knowledge is rejected or underutilized despite being of considerable potential value
26
27 (Kathoefer & Leker, 2012; Lichtenthaler & Ernst, 2006).
28
29
30
31
32
33
34

35 When NIH hinders the reception of knowledge, negative consequences are likely to occur. There
36
37 are many accounts of closely knit in-groups within companies considering their "insider"
38
39 knowledge as superior to outside knowledge. Apple Computer had such a mindset in the early
40
41 1990s, when managers rejected good external ideas and lived in what was widely known as their
42
43 own “reality distortion field” (Burrows 2000: 102; as quoted in Menon and Pfeffer, 2003: 497). A
44
45 striking instance of this kind of resistance was evidenced at Kodak over the last two decades,
46
47 where the company’s refusal to address the rise of digital over film technology led to its decline
48
49 and eventual bankruptcy (Lucas & Goh, 2009) . When Kodak developed the first digital camera
50
51 in 1975, the product was dropped for fear that it would threaten Kodak's photographic film
52
53 business. Later, in the 1990s, Kodak's top leadership developed a strong vision of a digital
54
55
56
57
58
59
60

company, but this strategy was scarcely implemented in its business units. Kodak had a long tradition of valuing consensus, in addition to a focus on internal development tracks. As a result, most of the people responsible for implementing the digital strategy had a background in film - a physical, chemical product dominated by the old Kodak. Employees knowledgeable about digital photography tended to be new employees hired to create change, but traditional managers were not quick to respond to the input and suggestions offered by their new colleagues.

Surprisingly, the popularity of the term NIH in the management literature has not been matched by a corresponding body of research (Agrawal, Cockburn, & Rosell, 2010; Kathoefer & Leker, 2012; Laursen & Salter, 2006; Lichtenthaler & Ernst, 2006). NIH has chiefly been studied in the context of innovation management, but has also featured in works on marketing (Franke, Keinz, & Steger, 2009; Hauser, Tellis, & Griffin, 2006), strategy (Fey & Furu, 2008; Laursen & Salter, 2006), or human resource management (McKinlay & Starkey, 1992). However, in most of this literature the underlying mechanisms of NIH are described in a rather anecdotal fashion (Agrawal et al., 2010; Kathoefer & Leker, 2012; Laursen & Salter, 2006). A systematic conceptualization, operationalization, and investigation of NIH is still missing.

The objective of our paper is to open the black box of NIH by providing an in-depth analysis of this frequently mentioned, yet rarely understood phenomenon. In doing so, we unearth the range of meanings assigned to the concept in extant research and identify those areas of literature that draw most strongly on the NIH concept. We review the causes and consequences of these barriers to external knowledge in the context of innovation and change. Complementing the vast literature on organizational inertia and tendencies to innovate in the known, we examine the NIH syndrome as rooted in individual attitudes. Importantly, NIH differs from organizational culture (Klein & Sorra, 1996). While organizational culture may act as a contextual factor influencing the

behavioral consequences of NIH, it is not the main source of NIH. As we will show, with reference to a large body of research in social psychology, NIH is caused by individual attitudes and their functions. Grounded in this extensive review we develop a definition of NIH. We offer a deeper understanding of the underlying causes of NIH, drawing on several theoretical perspectives with a clear focus on those attitudes known to shape individual thinking and decision-making. We integrate these different perspectives into a novel conceptual process model of NIH. Developing the existing research on NIH, we explicitly differentiate between behavior and the reasons (attitudes) that cause behavioral responses like underutilization or rejection of external knowledge. We conclude by identifying theoretical and managerial implications as well as opportunities for future research.

Conceptual Background

NIH has two key features. Firstly, it emerges as a result of boundaries that knowledge has to bridge when it is transferred between two entities. It therefore derives from some form of knowledge externality. Secondly, it involves an irrational devaluation or even rejection of this external knowledge by an individual, even though this knowledge might have been valuable from the perspective of the organization. In the following sections, we will further discuss these elements.

Sources of Knowledge Externality

Following Shannon and Weaver's (1964) basic model, knowledge is transmitted from a source to a recipient, either of which can be an individual, a team, or an organization. In this model, externality can be conceptualized in two different ways: externality with regard to the (disciplinary) context of the knowledge, and externality with regard to the source from which the

knowledge originates. The latter can be differentiated between organizational (functional) boundaries that the knowledge has to cross, and the spatial (geographic) distance between the knowledge source and its receiver. This differentiation results in the structure of externality presented in Figure 1.

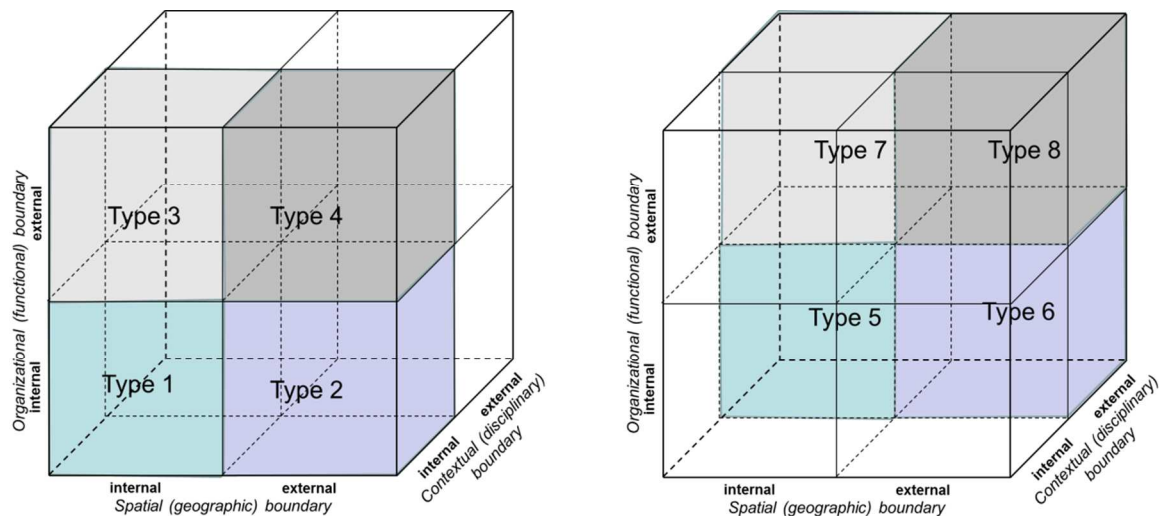


Figure 1: Dimensions of externality of knowledge leading to NIH

With regard to the *source of knowledge*, organizational boundaries can determine whether knowledge is perceived as external by an individual. Within an organization, knowledge can be transferred *intra-functionally* between persons from different teams (e.g. Bstieler & Hemmert, 2010) and *inter-functionally* between employees working for different functions, departments, or divisions of an organization (e.g. Song & Swink, 2009). Knowledge transferred between different organizations or institutions (e.g. Cassiman & Veugelers, 2006; Laursen & Salter, 2006) is regularly categorized as external knowledge. Furthermore, both internal and external organizational boundaries might coincide with a growing *spatial distance* between the knowledge

1
2
3 source and the receiver (Salomo, Keinschmidt, & De Brentani, 2010; Song, Asakawa, & Chu,
4
5 2011).
6
7

8
9 In Figure 1, only the field labeled "Type 1" represents a knowledge transfer which we
10
11 conceptualize as entirely "internal" and hence not potentially influenced by NIH. An example
12
13 would be a team member transferring knowledge to another team member located in the same
14
15 geographical location, with both colleagues having the same educational background, where this
16
17 background itself is the same as the discipline from which the knowledge is derived. NIH may
18
19 already occur when one of the team members is positioned in another area of the firm (Type 2),
20
21 even when she actually belongs to the same organizational unit. NIH may also occur when
22
23 knowledge has to cross hierarchical boundaries, as illustrated by Type 3. A famous example here
24
25 is Morrison's (1966) account of NIH in the U.S. Navy. When a junior officer pioneered an
26
27 innovation that dramatically increased the accuracy of firing guns from ships, senior naval
28
29 officers rejected it for many decades, only finally relinquishing when President Roosevelt
30
31 intervened.
32
33
34
35
36

37
38 Types 3 and 4 (as well as Types 7 and 8) describe the form of NIH that looms largest in the
39
40 literature: knowledge that is rejected as it stems from a different function or an external
41
42 organizational entity (firm). This was the mindset at Apple Computer in the early 1990s, where
43
44 managers were suspicious of strong external ideas (Burrows, 2000). For example, long after
45
46 market feedback and usability research indicated that the optimum number of buttons for a
47
48 computer mouse was two, Apple refused to modify their thinking and maintained its one-button
49
50 mouse design (Lidwell, Holden, & Butler, 2010). Resistance towards external knowledge is
51
52 typically assumed to increase with growing spatial distance. A questionable redesign occurred
53
54 with the American introduction of the British-made Sinclair Spectrum home computer, which its
55
56
57
58
59
60

US distributor Timex reengineered and released as the 'Timex 2068'. The changes increased the computer's capability, but the cost were a bland form factor and numerous software incompatibilities, resulting in the product's failure (Lidwell et al., 2010).

From a *contextual perspective*, knowledge can be perceived as external when it originates from another scientific or technical domain of expertise (Types 5, 6, 7, and 8 in Figure 1). Traditionally, scientific research is structured along disciplines. This division of scientific labor is mainly due to the reduction in complexity for the individual when concentrating on one specific aspect of the research problem at hand (Pierce, 1991). Disciplinary division helps to organize knowledge and leads to specialization, but also results in disconnected knowledge silos (Bitner & Brown, 2008). To address complex problems, however, research often needs to integrate knowledge from different domains within interdisciplinary projects (Jacobs & Frickel, 2009). Studies of knowledge transfer across disciplines reveal notable barriers to communication (Biehl, Kim, & Wade, 2006). Consider an example representing Type 5 in Figure 1. NIH often occurs between two team members in the same location, but with different disciplinary backgrounds. A lead engineer with a background in electrical engineering may reject a contribution based on knowledge stemming from mechanical engineering. In history, even outstanding scientists following different scientific ideologies (Field 8 in Figure 1) have rejected each other's ideas and theories (Nickerson, 1998), e.g. Leibniz and Huygens rejected Newton's concept of universal gravity due to their distinct scientific philosophies, even calling the concept occult (Lakatos, 1978).

NIH as an Attitude-Based Bias

Next to knowledge externality, the second element constituting NIH is an irrational rejection of knowledge in a case where this knowledge would actually have been valuable to the organization.

As we will discuss below, this can be the result of either a deliberate decision or of an unconscious, almost automatic rejection. NIH often originates from a different perception of rationality at the level of the organization and at the level of an individual. The person making the decision may act entirely rationally from an individual perspective - but this could just as easily be suboptimal for the organization. It is important to note, though, that rejection of external knowledge is not necessarily irrational. Rejecting external ideas or technologies when their expected value is outnumbered by costs, for example by transaction costs incurred when acquiring the knowledge but also by the cost of the knowledge stock itself (e.g. licensing fees), is fully rational. However, from the point of view of the organization, this evaluation should be rational, i.e. following the principle of value maximization. We hence need to distinguish NIH from an objectively rational rejection of external knowledge due to its lack of value or its high cost, but also from situations where external knowledge is rejected as actors just decide randomly (Sadler-Smith & Shefy, 2004) or engage in intended sabotage (Harbring & Irlenbusch, 2011). Both of these latter scenarios do not constitute NIH in our understanding.

NIH often resembles a principal-agent-setting where individuals have leeway when it comes to decision-making and therefore act opportunistically (Eisenhardt, 1989). In this context, we have to distinguish between an irrational, but conscious and deliberate, rejection and a reflex, unconscious rejection of external knowledge. Consider the example of ideas for product improvement provided by customers. While some of these ideas may be technically unfeasible or irrelevant for the target market (and hence it is fully rational to reject them), some ideas may indeed provide an opportunity to improve product performance. Nonetheless an individual developer may still reject even the most promising ideas in order to avoid adding a new burden to her workload, or simply as she characterizes the wording and language in which the idea has been

presented as "unprofessional" or "nerdy". Similarly, Gupta and Govindarajan (2000) highlight the fact that managers sometimes try to demote the power of peer units in an organization by portraying these units' knowledge stocks as less unique and valuable in order to protect their own decision power.

These examples are all characterized by an individual attempt to defend self-perception, rank or status. According to research in social psychology, these defense mechanisms are major functions of individual attitudes (Ajzen, 2001; Bohnert & Wänke, 2002). In general, attitudes are seen as a key determinant in influencing social life and human interaction (Eagly & Chaiken, 1993). Attitudes are defined as relatively time-consistent, individual evaluations of an object of thought, including physical artefacts, people, groups, and ideas (Bohner & Dickel, 2011). Research on discrimination (Agerström & Rooth, 2011; Toosi, Ambady, Babbitt, & Sommers, 2012) indicates that attitudes are often formed towards persons or groups that are perceived as external or foreign. In the context of NIH, however, the attitude object is the stock of knowledge that is perceived as external.

In accordance with prior research on NIH (Clagett, 1967; Kathoefer & Leker, 2012; Katz & Allen, 1982; Lichtenthaler & Ernst, 2006), we define NIH as a *bias triggered by the negatively-shaped attitude of an individual towards knowledge that has to cross a contextual (disciplinary), spatial or organizational (functional) boundary, resulting in either its sub-optimal utilization or its rejection as behavioral consequences of this attitude bias*. This understanding is also implied by calling NIH a "syndrome". In a decision-making context, the bias causes either a depreciation of the utility of the outside knowledge or an overestimation of the respective costs in obtaining

it.¹ For an innovating organization, this bias becomes economically negative when knowledge useful for the organization or its innovation behavior is rejected or underutilized.

Literature Review

To assess the current state of research, we conducted a detailed review of publications referring to the NIH syndrome based on an original set of 674 full papers referring to NIH.² We followed the approach of Lane et al. (2006), classifying these articles in a focused and a non-focused category according to the centrality of NIH to the paper's core topic. A NIH-focused paper examines the NIH syndrome itself, i.e. seeks to explain its occurrence, antecedents, or consequences. In contrast, a non-focused paper uses NIH to either explain an empirical result or

¹ Consider the example of a product developer rejecting promising ideas from customers, which could result from two process modes described in earlier literatures on decision-making: a more reflective or systematic approach and an impulsive or heuristic mode (Chen & Chaiken, 1999; Fazio, 1990; Strack, Werth, & Deutsch, 2006). The more deliberate mode is characterized as analytic, reflective, inclusive of context, and slow in processing. The more automatic approach is seen as emotional, impulsive and heuristic, and is characterized as conducive to swift decisions. Whenever decisions are led by heuristics, they might be biased (Gigerenzer & Gaissmaier, 2011). NIH predominately may be placed into this second, more impulsive mode of decision-making. Impulsive or heuristic decision-making is often driven by attitudes (Fazio, 1990; Strack et al., 2006). But NIH also can be seen as the result of a reflective, systematic rejection. In our example, the rejection of an idea because of its "nerdy" language or the general incentive system of a firm only honoring its "own" ideas can be seen as an automatic, impulsive decision. Not mentioning a good idea from customers out of fear that superiors may criticize the employee for "not having this obvious idea earlier" may be seen as a more reflective and systematic rejection.

² We started our review process by searching for relevant studies in ABI/INFORM, Business Source Premier, EconLit, Web of Science, and ScienceDirect. We used keywords such as "*not-invented-here*", "*not-invented-here-syndrome*", "*NIH*" and "*NIH-syndrome*" to scan the most informative text available in each database (headline, abstract, full text).

to derive hypotheses in another context. Therefore, we evaluated whether (1) NIH is used in the introduction to address a specific topic, (2) NIH is instrumental for the model, hypothesis, or proposition development, (3) the results are discussed or justified using NIH, or (4) the paper's outlook and limitations refer to NIH. In these four cases of instrumental or minor use of NIH, the article was categorized as “non-focused”, taking NIH for granted. Those papers analyzing NIH as a phenomenon in its own right (“NIH-focused”) were classified into three categories according to their research design: (1) qualitative research using case studies or ethnographic approaches (2) quantitative research using surveys, secondary databases or experiments, and (3) conceptual or theory-building articles.

Table 1: Analysis of the application of the NIH construct in management literature

Centrality of NIH		Research Design			Sum (percentage)
		Qualitative research	Quantitative research	Conceptual, research	
focused		2	10	2	14 (2%)
non-focused		187	169	304	660 (98%)
Number of non-focused papers using NIH in/for ...	introduction and motivation	29	20	28	77 (11.7%)
	hypotheses, model or proposition development	107	105	181	393 (59.5%)

result and discussion	48	46	75	169 (25.6%)
outlook and limitations	22	17	34	73 (11.1%)

The results of this literature review (Table 1) provide several unexpected insights. The analysis of the centrality of NIH to the paper's core topic reveals that 97.9 percent (660 papers) use NIH as a minor construct only. More than half of these articles (393 articles, 59.5 percent) use the construct during the development of the research model (hypotheses or propositions). Another 169 papers (25.6 percent) draw on NIH to discuss results: in quantitative studies often to justify unsupported hypotheses; in qualitative studies mostly to explain routines, decision-making processes, or behavioral effects. Less frequently but still often, NIH is used merely to introduce a study and address a research question (77 papers, 11.7 percent) or to discuss limitations or further research (73 articles, 11.1 percent). Conversely, only 14 of the publications identified (2 percent) explicitly investigate NIH and seek to deepen our understanding of it. We further analyzed the references cited for NIH in every paper. Interestingly, in the majority of papers, NIH is used as an established, concrete term, not as an acronym for a loosely-defined concept whose content might be up for analysis or discussion. About 65 percent of all papers do not cite any literature at all when referring to NIH. Surprisingly, the citations in the remaining 237 papers (35.9 percent) point to 67 different articles. The paper by Katz and Allen (1982) is by far the most central for the NIH literature stream, as it receives the maximum number of citations for one specific article (139). The mean number of citations an article gets is 3.5 (standard deviation 16.7) and the median is 1. Six further references received citations above this mean (Allen, 1977; Cohen &

Levinthal, 1990; Hayes & Clark, 1985; Jain & Triandis, 1990; Lichtenthaler & Ernst, 2006; Szulanski, 1996). Interestingly, from these seven papers, only Katz and Allen (1982) and Lichtenthaler and Ernst (2006) were classified as "NIH focused" in our analysis.

In conclusion, our analysis of the NIH literature indicates that NIH has become a taken-for-granted construct (Green, 2004). As Agrawal et al. (2010) put it, evidence for NIH is primarily anecdotal. NIH is frequently cited but it appears in what might be called a "ritual" way, providing little to no discussion of the phenomenon itself. In cases where such ritual citations occur, the perception of having tested assumptions and generated knowledge about the construct is often overestimated (Lane et al., 2006). Interestingly, NIH is one of the few academic concepts that are widely known and accepted in managerial practice. Ask any manager about problems of knowledge transfer and her answer will most likely include "not-invented-here".

However, given this broad exposure and common understanding of the concept, we were surprised to find only fourteen studies which focused on NIH in detail (refer to Appendix 1 for a detailed overview of the NIH papers). These papers uncovered a variety of antecedents to NIH, including established routines (Kathoefer & Leker, 2012; Katz & Allen, 1982), social identities of groups and organizations (Agrawal et al., 2010; Allen, Katz, Grady, & Slavin, 1988; Mehrwald, 1999), the human tendency to strive for security and stability (Kathoefer & Leker, 2012), poorly balanced incentive systems (Mehrwald, 1999), culture (e.g. individualism vs. collectivism) (Albach, Pay, & Rojas, 1991; Pay, 1989), and resistance to change (Clagett, 1967). As for its consequences, NIH may lead to incorrect evaluations of ideas and technologies (Agrawal et al., 2010; Burcharth et al., 2014; Kathoefer & Leker, 2012; Mehrwald, 1999), impeded implementation and increased development costs (Clagett, 1967; Pay, 1989), project

1
2
3 failure (Clagett, 1967; Herzog & Leker, 2010; Kathoefer & Leker, 2012), and a decline in firm
4
5 performance (Allen et al., 1988; Katz & Allen, 1982).
6
7

8
9 Prior research has used diverse measures to explain NIH. In most studies, NIH is not analyzed
10
11 directly. These studies interpret specific effects as potential NIH outcomes and use them as a
12
13 behavioral indicator of the underlying NIH attitude. The most cited NIH study, by Katz and Allen
14
15 (1982), measures the frequency of communication with external persons or institutions over time.
16
17 A decline in communication is interpreted as resulting from an increasing tendency to reject
18
19 external knowledge. In a recent study by Agrawal et al. (2010), NIH was operationalized through
20
21 self-citation rates in a patent analysis. The authors found that large and geographically isolated
22
23 firms tend to cite themselves more frequently. Mehrwald (1999) was first to develop a dedicated
24
25 operationalization, proposing a multi-dimensional concept to measure NIH that also takes into
26
27 account the theoretical definition of NIH as an attitude. Some recent studies have used single
28
29 factors and items of this operationalization to measure NIH (Herzog & Leker, 2010; Kathoefer &
30
31 Leker, 2012; Kathoefer, 2012). Recently, Burcharth et al. (2014) extended the attempts to
32
33 measure NIH by suggesting a third-person approach. R&D managers were asked to report on the
34
35 average attitude of employees towards external knowledge acquisition. This average measure,
36
37 however, neglects individual attitude differences and single behavioral consequences. Hence, one
38
39 can argue that this measure reflects more of a team climate than individual attitudes.
40
41
42
43
44
45
46

47 In conclusion, our literature review confirms that the existence of NIH is widely acknowledged.
48
49 However, only a limited number of studies have identified and discussed potential antecedents
50
51 and consequences of NIH. A few quantitative studies have started to take the original
52
53 conceptualization and definition as an attitude into account and developed tools to actually
54
55 measure NIH in an organization. Due to the rather limited number of evidence-based studies, the
56
57
58
59
60

existing literature still lacks confirmation of the attitudinal consequences and negative outcomes that it postulates.

Different Perspectives for Explaining the Bias Underlying NIH

NIH has been defined as an attitude-based bias against external knowledge. In order to enhance our understanding of this phenomenon, this section will discuss different ways in which this attitude may lead to biases in decision-making. In general, attitudes are seen as a main determinant influencing social life and human interaction (Eagly & Chaiken, 1993) through an evaluation typically encapsulating attribute dimensions like good-bad, pleasant-unpleasant, or likable-dislikable (Ajzen, 2001).

According to most attitude theorists, attitudes are seen as having five major functions. (Ajzen, 2001; Eagly & Chaiken, 1993) For an individual, an attitude helps to foster and maintain self-worth and self-concept (*ego-defensive function*) through expressing, activating and affirming a person's central values (*value-expressive function*). Hence, attitudes are part of a person's social identity (*social-adjustive function*). Furthermore, attitudes provide individuals with a simple structure and thus help to organize the complex and ambiguous environment (*knowledge function*). In a given situation, the perception of an attitude-object as good or bad helps individuals to evaluate the object's use when it comes to an approach or avoidance decision (*utilitarian function*) (Bohner & Wänke, 2002; Pratkanis & Greenwald, 1989). Interestingly, identical attitudes may fulfill different functions for different people and even more than one function for a single individual. This implies that an attitude may be multifunctional (Bohner & Wänke, 2002).

The different functions of attitudes influence the extent to which individuals (consciously) process attitude-consistent or inconsistent information (Petty & Wegener, 1998). Individuals often seek and select information that confirms and supports their personal beliefs and attitudes (Munro & Ditto, 1997). At the individual level, attitudes guide thinking, decisions, and behavior in a heuristic way and are likely to lead to biased decision-making (Ajzen, 2001). At the social level, intergroup exchange, cooperation and conflicts are often influenced and induced through attitudes towards both one's own group and external groups (Bohner & Wänke, 2002). In the following, we will apply the five functions of an attitude to elaborate several theoretical perspectives encompassing an attitude as a main factor driving individual thinking and decision-making in the context of NIH, referring to the different levels of externality of knowledge as the origin of NIH.

The ego-defensive function of NIH

Gupta and Govindarajan (2000) argue that an ego-defense mechanism may underlie NIH. Individuals might block information proving or suggesting that others are more competent than they perceive themselves. Typically, ego-defense mechanisms include denial, projection, repression or rejection to shelter an individual's self-concept (Eagly & Chaiken, 1993). In a corporate context, managers and developers often define and express their self-identity through their expertise in a specific domain (Menon, Thompson, & Choi, 2006). This feeling of possession can lead to a kind of "psychological ownership" of this specific problem or knowledge domain, thus fostering the expectation of having responsibility for solving the problem in question (Pierce, Kostova, & Dirks, 2001). Challenging this self-perception might result in impediments, failure to share information, sabotage or other deviant behavior, and may lead to stress, frustration, and other psychological effects (Pierce et al., 2001). Concerning knowledge

transfer, Baer and Brown (2012) have shown that perceiving psychological ownership of ideas leads to the rejection of suggestions by others in this area. Along with psychological ownership, ego-defense might also link with the self-serving bias (Kelley, 1967). Research has broadly discussed an individual's tendency to overestimate their own contributions and attribute failure to external factors which influence decisions or tasks (Cucina, Martin, Vasilopoulos, & Thibodeaux, 2012; Nelson & Beggan, 2004). Hölzl, Kirchler, and Rodler (2002) have shown that such self-serving behavior is routed in an individual's attitude.

Applied to the context of knowledge transfer, showing a tendency to NIH may be triggered by the ego-defensive function of the underlying attitude. In an R&D context, researchers will often have been hired because of their unique expertise in a specific domain. Expert tracks, the notion of "innovation champions", or an award culture recognizing expert achievements (like Procter & Gamble's famed Victor Mills Society) foster the construction of pockets of expertise obtained by one individual. Such an individual may demonstrate NIH when confronted with external knowledge in this particular domain in order to defend her expert status.

The value-expressive function of NIH

Attitudes are also a means of expressing individual values and ideologies (Bohner & Wänke, 2002). The value-expressive (also: ego-expressive) function of an attitude helps people to satisfy the need to clarify and confirm their self-concepts, showing what kind of individual they are and what they stand for (Eagly & Chaiken, 1993). According to Klein and Sorra (1996), employees are more willing to accept, adopt, and implement innovations when they are consistent with both individual and organizational values. For example, a conventionally educated pharmacologist or medical doctor may deny effects reported as the result of homeopathic remedies, viewing them as a placebo rather than a medicinal effect. Parents have been shown to decide not to vaccinate their

children because of the potential chances of death as a rare side-effect of the medication, even when the probability the vaccination will cause death is much lower than the probability of dying from the disease (Ritov & Baron, 1990). The value function of the parents' attitude (the desire to protect their child's life) biases the rational decision towards making the wrong choice. It also is very likely that an environmentalist will reject product ideas that contain potentially toxic materials or require air-polluting production processes, even if the products could increase customer welfare. Thus, drawing self-esteem from following a specific ideology or value and having related attitudes is expected to lead to a strong rejection of other ideologies, schools of thought, or values as embodied in external knowledge inputs.

The social-adjustive function of NIH

Attitudes serving the social-adjustive function influence a person's relation towards others. These attitudes can facilitate and maintain social relationships, but at the same time can also disrupt and violate them (Eagly & Chaiken, 1993). Hence, attitudes influence the development of a social identity (Bohner & Wänke, 2002). Social identity theory postulates that part of a person's self-concept is derived from membership of social groups (Tajfel & Turner, 1979). A positive social identity thus helps to satisfy the need for a positive self-esteem (Hewstone, Rubin, & Willis, 2002). While the ego-defensive function of an attitude helps to shelter the individual part of the self-concept, the social-adjustive function can be seen as protecting, and also fostering, the group-related part of the self-concept. Such an attitude, for example, can lead to degrading others as a means of elevating one's individual self-concept, conforming to group norms, or justifying existing structures and power differentials (Bohner & Wänke, 2002). Research has shown that the social identity of a group and induced group cohesion can enhance the performance of a team

(Scott, 1997). Exchange and cooperation between teams and groups having different social identities, in contrast, can be impeded (Ashforth & Mael, 1989; Skilton & Dooley, 2010). Social identity theory has frequently been used to explain why NIH exists and which consequences it might entail. Research studying multinational organizations has argued that spatially distributed teams develop separate social identities which hamper the exchange of knowledge and intra-organizational cooperation, as strong local social identities lead to the rejection of external knowledge, even if it comes from members of the same organization (e.g. Chang, Gong, & Peng, 2012; Haas, 2010; Kostova, 1999). Moreover, a strong team identity might also foster inter-organizational NIH (Katz & Allen, 1982; Menon & Pfeffer, 2003). Other studies have discussed country-specific cultures and resulting identities (Michailova & Husted, 2003) to explain NIH occurring between organizations. Moreover, the social identity of an organization as a whole (corporate identity) may lead to the rejection of external knowledge (Agrawal et al., 2010; Husted & Michailova, 2002; McEvily, Perrone, & Zaheer, 2003). This pattern has, for example, recently been shown in German companies where a strong "German engineering pride" prevents the absorption of relevant external solutions in a R&D context (Lüttgens, Antons, Pollok, & Piller, 2014). Similarly, Kodak's failure to alter its major technology from analog films to digital photography is partially seen as a result of employees' rigid mindset to preserve the status-quo, thereby avoiding any disturbance of the social harmony within the R&D team through external innovation (Lucas & Goh, 2009).

The knowledge function of NIH

Attitudes provide individuals with a simple structure to organize the processing of knowledge. This function is one of the most prominently discussed functions of an attitude in the literature. Attitudes help an individual to simplify and organize the perception and evaluation of

1
2
3 information in a situation characterized by complexity and ambiguity. The knowledge function
4
5 supports individuals in attaining a meaningful, systemized and stable perspective (Eagly &
6
7 Chaiken, 1993). Striving for cognitive consistency, people filter out new information that
8
9 challenges their attitudes and adopt information that is in line with their attitudes (Ajzen, 2001;
10
11 Bohnert & Wänke, 2002). This attitude-induced selective information processing affects attention,
12
13 encoding, exposure, judgment, elaboration and information storage (Bohner & Wänke, 2002;
14
15 Eagly & Chaiken, 1993).
16
17
18

19
20 In the context of NIH, the knowledge function of an attitude can lead to a confirmation bias
21
22 (Nickerson, 1998). It has been shown that scholars reviewing articles tend to prefer submissions
23
24 that do not deviate from their specific expertise or the common prevailing wisdom (Hergovich,
25
26 Schott, & Burger, 2010). As Nickerson (1998) shows with a notable collection of examples,
27
28 famous scientists, including Galileo, Kelvin, Davy, Huygens, and Leibniz, declined ideas and
29
30 theories offered by upcoming scientific contemporaries (all of whom later also became famous
31
32 for their contributions), and instead continued to rely on their professional knowledge and
33
34 perception of the field. In organizations product managers and developers tend to rely on their
35
36 initial positive assessment of the market potential for a novel product, even when contrary
37
38 information later becomes available (Biyalogorsky, Boulding, & Staelin, 2006). Similarly, it has
39
40 been reported that decision-makers often reject developments by innovative users (von Hippel,
41
42 1988), as these lead users often address usage situations of a product which are different to those
43
44 envisioned by management. The literature on lead users contains countless examples of such
45
46 rejections. Von Hippel (2005), for instance, mentions the examples of the invention of white-out
47
48
49
50
51
52
53
54
55
56
57
58
59
60

liquid or the "CamelBack", a hydration device for sportsmen. Both inventions were repeatedly rejected by established vendors in the respective industries.³

The utilitarian function of NIH

People generally seek to gain rewards and avoid punishments. Attitudes fulfilling the utilitarian function support the securing of positive outcomes and the prevention of negative ones from the perspective of an individual (Bohner & Dickel, 2011; Demski & McGlynn, 1999). In general terms, utilitarian attitudes exemplify individual self-interests (Eagly & Chaiken, 1993). In organizations, employees are often incentivized to develop new ideas or advance technological solutions leading to a patent application (Im, Montoya, & Workman, 2013). Such incentive and reward systems, however, may have unintended effects on creativity and ideation (Koch & Leitner, 2008). Consider developers or product managers who are rewarded for the number of ideas or development projects generated. In such a situation, the rejection of external ideas may occur due to the sheer utilitarian function of NIH, securing positive outcomes for the respective developer (Argote, McEvily, & Reagans, 2003; Hauser et al., 2006; Menon & Pfeffer, 2003). Besides monetary rewards, creating one's own ideas may also simply be more prestigious than adapting an external idea, as it fosters intrinsic motivation via peer recognition or social status in an organization (Husted & Michailova, 2002). Hence, the utilitarian function of an attitude can lead to an ownership bias or endowment effect in idea generation and evaluation, where

³ A recent account of this effect has been the initial rejection of the idea by an early Twitter user to use the hashtag symbol (#) to channel tweets about a specific event. The Twitter development team considered the hashtag as being "too nerdy" and not appealing to a wider audience, and hence did not promote the idea (Zak, 2013). Twitter only changed its policy after users continued to use the symbol. Today, hashtags are one of the most ubiquitous means of structuring communication in social media.

employees are more likely to promote their own ideas over external ideas (Onarheim & Christensen, 2012). This behavior has been revealed in the context of ideation contests, for example, where customers submit ideas to improve a firm's offerings. Even the winning ideas submitted to these contests are often later rejected by the hosting company, as its managers have no incentive to adopt them in their own projects.

Connecting the functions

Table 2 summarizes the set of functions developed above, explaining how attitudes trigger the rejection of external knowledge. Complementing the differentiation of three distinct sources of knowledge externality (organizational, spatial, and contextual boundaries) introduced earlier, this table can serve as a framework for a more finely grained description and investigation of the various instances of NIH.

Table 2: Behavioral trajectories of attitudinal functions

NIH attitude function	characterization	triggered heuristics and related theories	related literature (selected)
<i>ego-defensive function</i>	defining, expressing and defending self-identity	- psychological ownership - self-serving bias	Baer & Brown (2012); Gupta & Govindarajan (2000)
<i>value-expressive function</i>	clarifying and confirming self-concepts and values	omission bias	Ritov & Baron (1990)
<i>social-adjustive function</i>	facilitating and maintaining social relationships	social identity theory	Agrawal, Cockburn, & Rosell (2010); Chang, Gong, & Peng (2012); Haas (2010); Skilton & Dooley (2010)
<i>knowledge function</i>	providing simple structures to organize information processing	- cognitive consistency and selective information processing - confirmation bias	Biyalogorsky et al. (2006); Hergovich et al. (2010); Nickerson (1998)
<i>utilitarian function</i>	securing positive outcomes and preventing negative ones	- ownership bias - endowment effect	Argote, McEvily, & Reagans (2003); Hauser, Tellis, & Griffin (2006); Hauser (1998); Koch & Leitner (2008); Onarheim & Christensen (2012)

Obviously, the five functions of an attitude may strongly interact. Consider again Morrison's (1966) description of NIH in the US navy. The innovation reported by a junior officer was rejected by senior officers in Washington – an instance where knowledge had to cross hierarchical boundaries. It was only after personal intervention by President Roosevelt that the new ammunition technology was implemented navy-wide. According to Morrison (1966), the main reason for rejecting the new technology was that it criticized the gun machinery that was originally designed by the senior officers to whom the junior officer was sending it. At this point, different attitude functions like the ego-defensive and the utilitarian function triggered a deliberate rejection of the innovation. The senior officers had been part of the development of the established systems and the underlying routines, leading to their perception of being the major experts for gun fire at sea. They developed strong psychological ownership in the existing (internal) solution fostered by the ego-defensive function of NIH. In addition, their personal careers, and their future development, were dependent on their own expertise. Thus, the rejection might have also been caused by the utilitarian attitude function in order to justify their existing positions and benefit their future careers. At the same time, an omission bias may have led to an unconscious rejection of the idea that a mechanical apparatus could replace human intelligence in precise gun aiming (value-expressive function). The same unconscious rejection may have been caused by selective processing of information (knowledge function), for example by focusing on instances of failure of the new technology while suppressing all information demonstrating its higher accuracy.

One may argue that this is an old example in a hierarchical organization unable to change. Recent research in the context of open innovation, however, demonstrates similar behavior in organizations that implemented a dedicated program to source external knowledge. Sieg, Wallin,

and Krogh (2010) and Lüttgens et al. (2014) observed European and US manufacturers who actively implemented an open innovation program by contracting an intermediary (like NineSigma, InnoCentive, and Yet2) to solve an existing internal technical challenge (Afuah & Tucci, 2012; Jeppesen & Lakhani, 2010). These open innovation initiatives were all sponsored by the CTO of the respective organization, hence providing a situation where the firm had established a formal procedure to acquire external knowledge for a given innovation problem. Still, both studies find a strong rejection of external solutions in all of the observed open innovation projects . In the study by Lüttgens et al. (2014), not one of the firms acquired an external solution or engaged in further cooperation with a successful solution provider.

Our framework helps to better understand this resistance and to open the "black box" of NIH. The external solution knowledge from the solution provider had to cross spatial and contextual boundaries (Types 4 and 8 in the structure of externality presented in Figure 1). In particular the access to solution knowledge from different industries and disciplinary domains has been demonstrated as one of the main objectives of open innovation (Jeppesen & Lakhani, 2010). This was also the motivation of those organizations studied by Lüttgens et al. (2014) and Sieg et al. (2010). Hence, when it came to integrating the external solutions, team members identified several reasons for not utilizing the external knowledge input. Consider the following quotes from interviews documented in Lüttgens et al. (2014: 352f. and 356): *"I can't tell you what it's like to stand in front of conservative managers whose leadership has not been able to solve the problem for years, and then you suggest solving the problem, but now with the help of external parties."* and *"R&D says: Why should we find an external solution when we can do it ourselves? If we do this, we're really only showing how bad we are."* These quotes illustrate how the ego-defensive function interacts with the utilitarian function (e.g. endowment effect for internally

developed solutions). Other interviewees reported that this rejection was shared broadly among members of the R&D teams. This illustrates the social-adjustive function facilitating an internal bonding of different R&D members, unanimously rejecting the winning external proposals.

These examples demonstrate that a negative attitude towards external knowledge is often triggered by various attitude functions, which may show a reinforcing effect. While it is difficult to differentiate these effects empirically with the methods currently available, we believe that a theoretically-grounded distinction of the five functions provides a better understanding of the underlying causes of NIH to inform future research. In conclusion, Figure 2 introduces a process model which allows us to open the "black box" of NIH. Our model consists of three elements: the different sources of knowledge externalities constituting the decision setting, the inner structure of NIH based on the attitude functions, and the resulting behavioral effects and performance impacts. Extending the argumentation in the existing literature on NIH, we explicitly differentiate between behavior and the reasons (attitudes) that cause behavioral responses such as the underutilization or rejection of external knowledge. The individual decision regarding the utilization of knowledge originating from contexts beyond disciplinary, organizational, and spatial boundaries may be affected by attitudes towards this external knowledge. These attitudes and their functions can trigger different heuristics that may mediate the attitude's influence on decision-making. When the decision is biased negatively against utilization, we call this situation a NIH bias causing either a depreciation of the utility of the outside knowledge or an overestimation of the respective costs in obtaining it.

Extending the current discussion, our model also indicates that different contextual factors may influence the attitude-behavior relationship and its resulting performance impact. These factors include the organizational culture (Herzog & Leker, 2010; Kostova, 1999), the incentive system

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60

in place (Im et al., 2013), the commitment of top-management (Fey & Furu, 2008), as well as other factors influencing the absorptive capacity of an organization (Zahra & George, 2002). While previous literature has described the performance impact of these contextual factors in general, their influence on the link between an underlying attitude and a behavioral response has not been subject of research. However, as our examples and the theoretical discussion have shown, different attitudes tend to trigger different heuristics, which also demand different approaches to deal with them. Based on our conceptual model and its underlying theoretical considerations, we identify a number of theoretical and managerial implications, which hopefully will serve as starting points stimulating further research. These are presented in the following section.

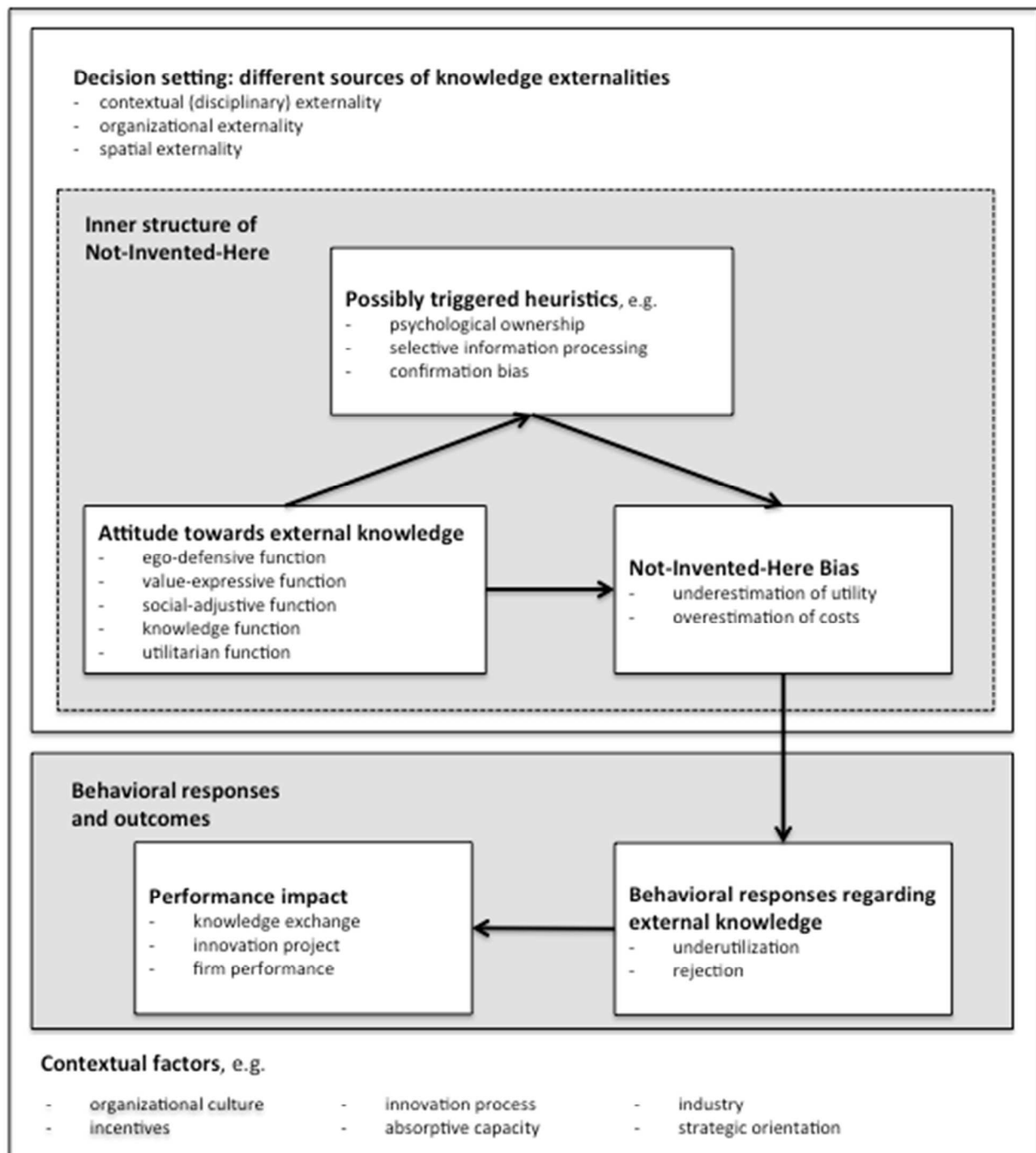


Figure 2: Conceptual model of NIH

Avenues for further research on NIH

Our literature review revealed that there is surprisingly little focused research on NIH. In the following, we apply the insights gained from both reviewing existing NIH literature and identifying the five attitude functions to develop promising avenues for future research on NIH. That said, we do not intend to develop a comprehensive list of research opportunities. Instead, we identify five broader issues that appear particularly critical for extending our understanding of NIH: (1) developing refined measurement instruments for NIH (2) studying the behavioral consequences of an NIH attitude (3) comparing NIH across contexts (4) developing and testing institutional remedies to overcome NIH, and (5) shifting the perspective from the individual to NIH in a group setting.

Measurement of NIH

First, research is required to develop refined measurement instruments for NIH in organizations. A validated instrument is not only crucial to investigate the extent of NIH in an organizational unit, but also to develop countermeasures and test their effect. Only very few studies have tried to provide such a NIH measure. In many instances, observed effects attributed to NIH such as deprecating behaviors are actually measured instead of NIH attitudes per se (Agrawal et al., 2010; Lichtenthaler & Ernst, 2006). The few papers (e.g. Herzog & Leker, 2010; Kathoefer & Leker, 2012) that provide a dedicated psychometric NIH measure focusing on the underlying attitude refer to Mehrwald (1999), who, to our knowledge, has provided the first direct measure of a NIH attitude relying on multiple Likert scale items. Applications of this measure have focused on demonstrating the existence of NIH rather than seeking to identify the antecedents and consequences of the phenomenon (Mehrwald, 1999). From a theoretical perspective, these measures can be criticized for neglecting established theoretical findings from psychology,

1
2
3 including most notably the threefold component structure of an attitude (Bohner & Wänke, 2002;
4
5 Eagly & Chaiken, 1993; Rosenberg & Hovland, 1960).
6
7

8
9 More importantly, the current measures of NIH in the management literature use direct items
10
11 within a standardized questionnaire containing questions about the utilization of external
12
13 knowledge, the collaboration with external partners, or trust in internally and externally
14
15 developed technology (Kathoefer & Leker, 2012; Mehrwald, 1999). Methodological literature in
16
17 general (Bryman, 2008) and psychological literature on attitude measurement in particular
18
19 (Ajzen, 2001; Fazio & Olson, 2003), however, criticizes such an explicit way of investigating an
20
21 individual attitude. Directly asking people about their attitudes can lead to response biases like a
22
23 social desirability response set. The more sensitive the specific attitude domain is, the greater the
24
25 likelihood that motivational issues occur, leading to a concealment of the real attitudes (Fazio &
26
27 Olson, 2003; Uhlmann et al., 2012). Similar to questions about racism or sexual orientation in a
28
29 workplace environment, we also believe that directly measuring NIH will be biased by social
30
31 desirability. In an area of open innovation, accepting external knowledge and ideas is proposed as
32
33 desirable, and hence respondents may not be willing to reveal their real attitude. In addition, an
34
35 attitude like NIH can lead to unconscious, almost automatic behavior, even in situations when an
36
37 individual would rationally perceive herself as open and professional.
38
39
40
41
42
43
44

45 To overcome these shortcomings, research in psychology has developed an abundance of so-
46
47 called indirect or implicit attitude measures (Fazio & Olson, 2003; Nosek, Hawkins, & Frazier,
48
49 2011; Uhlmann et al., 2012). Applying these measures to NIH will have the potential to broaden
50
51 the understanding of the underlying attitude but also to improve and validate direct attitude
52
53 measures. Indirect attitude measurement techniques both minimize the individual awareness of
54
55 the measured constructs and the ability to control and regulate responses. In most cases, non-
56
57
58
59
60

reactive measures like reaction times, picture or word classifications, and sentence completion are used. Applying such measures holds great promise for research on NIH where social-desirability might occur due to the aforementioned attitude functions and related effects.

Consequences of NIH

Second, research is needed to provide robust evidence of the expected causal link between the NIH attitude and suboptimal decision-making (see Figure 2). While many authors argue that NIH leads to project failure or underperformance, this causal claim has rarely been tested, which is not least due to the lack of appropriate measurement instruments. Further research should take the functions of an attitude developed in the previous section into account in order to understand how NIH influences behavior and induces a general rejection of external knowledge. Different research designs appear particularly promising in this regard.

In line with literature on innovation barriers (Klein & Sorra, 1996), we see a strong opportunity in using lab experiments to focus on the attitude-behavior relationship and to investigate how the attitude effect is potentially mediated through other biases. Furthermore, larger, more conventional surveys could be a good means of evaluating possible moderators of these relationships and to investigate the proposed trajectories linking attitudes and behavioral responses. We also see strong potential in applying qualitative case studies to observe managers and the formation of their decision-making in a firm context. Qualitative research could also investigate the origin of ideas and technological breakthroughs to better understand those factors that trigger resistance towards, or conversely adoption of, the underlying knowledge stocks in a particular setting. Research needs to apply the full range of methodological tools to understand the complex phenomenon of NIH and to differentiate the various behavioral consequences of an NIH attitude in different organizational settings.

Findings from such research could also help to solve a persisting theoretical and empirical puzzle in the open innovation literature (Gesing, Antons, Piening, Rese, & Salge, 2014). Research drawing on absorptive capacity theory (Cohen & Levinthal, 1990; Lane et al., 2006; Zahra & George, 2002) argues that internal R&D capabilities are prerequisites for attracting competent collaboration partners (Dahlander & Gann, 2010; Grimpe & Kaiser, 2010), resulting in complementarity of internal and external R&D. Other authors assume a substitution effect of external R&D collaboration due to perceiving it as questioning the legitimacy of internal R&D and thus leading to resistance of R&D staff (Laursen & Salter, 2006). Empirical research provides support for both complementarity (Cassiman & Veugelers, 2006; Grimpe & Kaiser, 2010) and substitution effects (Hess & Rothaermel, 2011; Laursen & Salter, 2006). Developing NIH measures might help to solve this puzzle by supplementing existing measures of absorptive capacity by instruments to evaluate NIH in order to investigate their opposing effects on open innovation performance.

Contexts of NIH

Third, after establishing a valid measurement instrument of NIH attitude and its consequences, NIH needs to be examined in different contexts. NIH was originally described as a phenomenon which arose from a collaborative innovation project (Clagett, 1967; Katz & Allen, 1982). Later, the idea of NIH was applied beyond innovation management, covering the entire domain of intra- or inter-organizational knowledge transfer (Szulanski, 1996). As discussed, NIH may occur along three boundaries: organizational, spatial, and contextual boundaries. Its original conceptualization by Clagett (1967) was in the context of knowledge crossing disciplinary boundaries. However, in most of the existing literature, NIH has only been applied to explain resistance towards knowledge from an external source, e.g. crossing organizational boundaries between firms

(Laursen & Salter, 2006) or spatial boundaries between teams of a multinational corporation located in different geographical regions and cultures (Gupta & Govindarajan, 2000; Kostova & Roth, 2002).

We see great opportunities for future research that also takes the content (discipline) of the knowledge being transferred into account. For instance, interdisciplinary teams and knowledge exchange have been shown to positively influence innovativeness and innovation performance (Talke, Salomo, & Rost, 2010). However, this effect diminishes with increasing cognitive distance, following an inverted U-shape (Wuyts, Colombo, Dutta, & Nooteboom, 2005). Future research should investigate whether this decrease could be explained by an increase of negative NIH attitudes resulting from the cognitive distance between different knowledge domains.

Countermeasures for NIH

Fourth, researchers need to identify and evaluate explicit countermeasures to overcome NIH. The literature has proposed various remedies such as rotating team members on a project basis, integrating employees into decision-making, re-structuring teams and departments, gaining experience with external knowledge, and establishing adequate incentive systems (Kathoefer & Leker, 2012; Katz & Allen, 1982; Lichtenthaler & Ernst, 2006). However, the effectiveness of these approaches has not yet been evaluated, not least due to the lack of appropriate measures.

We see two different types of mechanisms for overcoming NIH. Apparently, changing someone's attitude is one way to reduce NIH, i.e. creating an attitude that allows an individual to overcome intuitive resistance to external knowledge. Research on persuasion from social psychology views attitude change as the result of better information processing (Bohner & Wänke, 2002). To induce such processing, several approaches exist, ranging from conditioning individuals through

1
2
3 exposure to information, persuading them via socially-imparted or incentive-motivated
4
5 experience, or inducing positive, emotional or behavioral experiences (Bohner & Wänke, 2002;
6
7 Crano & Prislin, 2006; Eagly & Chaiken, 1993; Wood, 2000). These countermeasures are
8
9 illustrated in Figure 2 as contextual factors shaping individual decision-making. Specific means
10
11 include informing team members about the causes, costs, and remedies of NIH, as recognition is
12
13 the first step to prevention and recovery (Lidwell et al., 2010). Moreover, encouraging decision
14
15 makers to interact with a wider community of external actors on conferences or in cross-industry
16
17 work groups may prevent a negative attitude towards external knowledge. Future research should
18
19 study the effectiveness of these measures, also in connection to different sources of knowledge
20
21 externality.
22
23
24
25
26

27
28 An alternative, perhaps less obvious way to overcome NIH is to prevent the attitude from
29
30 influencing behavior (instead of changing the attitude). As described above, attitudes might act as
31
32 information processing heuristics biasing decisions. Recent literature has suggested so-called de-
33
34 biasing techniques which help individuals to avoid heuristics during decision-making (Milkman,
35
36 Chugh, & Bazerman, 2009). These techniques include analogical reasoning, the use of foreign
37
38 language to communicate during a knowledge exchange, or perspective-taking approaches
39
40 (Galinsky & Moskowitz, 2000; Keysar, Hayakawa, & An, 2012; Milkman et al., 2009).
41
42 Analyzing a bias and understanding the reason behind it is at the very core of the de-biasing
43
44 approaches. Consider the example of a cross-functional team including employees from
45
46 marketing, R&D and production. When marketing offers ideas based on their customer-related
47
48 knowledge that might enhance the functionality of a new product, R&D employees might reject
49
50 these ideas because of NIH-related ownership effects. To overcome this rejection, the project
51
52 leader could ask the developers to take a marketing perspective and try to understand why the
53
54
55
56
57
58
59
60

team proposed these ideas. Once this understanding has been obtained, a strategy to undermine the biasing mechanism can be developed (Milkman et al., 2009). The different functions of an NIH attitude, as discussed before, are a good starting point to analyze the respective de-biasing mechanisms. Assuming that in the given example ownership effects arise due to the ego-defensive attitude function, future research should validate that perspective-taking is indeed a reasonable countermeasure.

Levels of Analysis

Fifth, further research about NIH might wish to shift its perspective from the individual to the group level. Multilevel research distinguishes two fundamental processes describing how distinct organizational levels interact (Kozlowski, Chao, Grand, Braun, & Kuljanin, 2013): bottom-up and top-down. Bottom-up emergence resides in lower-level entities such as the individual and individual effects and habits arise to upper levels. Over time, these emergent phenomena then manifest themselves at higher and more collective levels, for instance in the form of a specific organizational culture. Top-down effects comprehend higher-level phenomena which shape, impact, or constrain lower-level phenomena. These effects include creating an open-minded corporate culture, implementing incentive systems, and trying out different ways of personal leadership (Hunter, Perry, & Currall, 2011). In general, the phenomenon of emergence has largely been neglected in quantitative research in the context of knowledge transfer in general and NIH in particular, but also has been sparsely covered in qualitative studies (Kozlowski et al., 2013). We suggest that studying organizations and individuals over time is crucial to understanding the evolution of NIH. A possible means of doing so would be to study groups where specific individuals show NIH tendencies. Does NIH then also emerge at the group-level? What moderates this process? Is it, for example, immanent personal traits, or capabilities like

leadership of individuals demonstrating NIH tendencies? Research answering these and other multilevel and longitudinal questions will foster our understanding of NIH and multilevel interaction effects.

Implications and Outlook

The NIH syndrome is one of the most frequently mentioned phenomena in the current era of open innovation and interdisciplinary knowledge exchange. As shown in our literature review, however, there is a rather fuzzy understanding of NIH, its antecedents, underlying attitudes, and behavioral consequences. The objective of our paper has been to provide a much needed definition and conceptual understanding, grounded in recent research in psychology and an extensive review of the management literature on NIH. We developed a framework of different sources classifying knowledge as "external" which might trigger a general, attitude-based rejection of such knowledge inputs irrespective of their specific value. Differentiating various functions of an attitude, we identified possible trajectories linking NIH with individual behavior and decision-making..

We then applied this understanding to develop an extensive agenda for future research. A core element of this agenda consists in the development of more appropriate instruments to measure the NIH attitude of an individual without inducing a bias of social-desirable answering behavior. Transferring recent research in social psychology on implicit measurements may be one fruitful avenue in this regard. This shall allow further research to develop measures and then test organizational interventions to face NIH. In so doing, we complement the literature on organizational measures and structural responses to foster innovation at the organizational level by adding an individual level perspective on the rigidity of knowledge transfer for innovation.

Here our model is in line with recent research (Rogan & Mors, 2014) on the micro-foundations of organizational ambidexterity that studies how broadly managers search for external information to explain innovation performance.

We have defined NIH as a negative attitude. Intuitively, individuals will differ in the extent to which they hold this attitude. Some research has even demonstrated a positive bias towards external knowledge (Laden, 1996; Menon & Pfeffer, 2003). While we have followed the dominant conception of NIH in the literature as a negative attitude, future research may broaden the debate on attitudinal effects on knowledge transfer and investigate positive biases stimulated by our theorizing. For example, Laden (1996) has described a kind of buy-in syndrome leading to an irrational preference for external input in a decision context.

NIH has been discussed predominantly in the context of innovation management and inter-organizational knowledge transfer. That said, two additional academic discourses may benefit from our research. Given that NIH is defined as the negative part of a mental attitude towards external knowledge, scholars interested in attitudinal effects or decision-making biases in organizational settings are expected to benefit from a better understanding of NIH and its measurement. This encompasses, for example, studies on discrimination effects in workplace environments. Up to now, such research looks into attitude effects towards persons or groups based on race (Toosi et al., 2012), gender (Elkins, Phillips, & Konopaske, 2002), or age (Wood, Wilkinson, & Harcourt, 2008), influencing hiring (Agerström & Rooth, 2011), salary negotiation (Jarell & Stanley, 2004), or promotion (Lyness & Heilman, 2006). Here, introducing NIH as an attitude-based effect towards knowledge into the discrimination literature might stimulate and broaden the debate on discrimination in the workplace, especially with regard to inventive activities.

1
2
3 Similarly, scholars of decision-making are interested in understanding effects and heuristics that
4 bias individual decisions because of bounded rationality (Simon, 1955). This literature covers a
5 broad variety of psychological tendencies and processes influencing individual decisions like the
6 recognition heuristic (Goldstein & Gigerenzer, 2002), the representativeness heuristic (Choliz,
7 2010; Read & Grushka-Cockayne, 2011), or anchoring effects (Englich, Mussweiler, & Strack,
8 2006). Introducing (attitudinal) phenomena discussed in social psychology to this field of
9 research might deepen the understanding of these heuristics, their antecedents and behavioral
10 trajectories. In specific situations attitudes might trigger heuristic decision-making while
11 fulfilling the functions discussed above for an individual.
12
13
14
15
16
17
18
19
20
21
22
23
24

25 We also see important implications for management practice. Given that NIH is a phenomenon
26 that is widely known (and bewailed) by managers, we believe that a better understanding of NIH
27 also contributes to better practice. First of all, we recommend that managers should acknowledge
28 that NIH exists – probably even in their organization. However, it would be mistaken to take NIH
29 for granted and use it as an easy excuse when open innovation initiatives fail or external ideas are
30 not utilized in a project. Procter & Gamble's famous expression of "proudly developed
31 elsewhere" (Elmquist, Fredberg, & Ollila, 2009) provides a cultural indicator of preferred
32 employee behavior. Such behavior, however, will only materialize if these claims are also
33 supplemented by corresponding incentives. Some of the organizations we studied have started to
34 build such explicit incentives to overcome NIH, like an award culture recognizing successful
35 transfer of external knowledge. Other organizations have started to engage outsiders in both the
36 strategy and the evaluation stages of the development process to ensure fresh perspectives and
37 new thinking. In one organization, we observed a "democratized" evaluation process where at
38 least 30 managers from different units and cultural backgrounds are involved in idea evaluation,
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60

1
2
3 hence counterbalancing individual NIH tendencies of a focused decision board. Clearly, there is a
4
5 compelling case for future research studying the effectiveness and efficiency of these and other
6
7 approaches to finally find a cure for the NIH syndrome.
8
9

10
11
12
13
14
15
16 **References**

17
18 Afuah, A., & Tucci, C. L. (2012). Crowdsourcing as a solution to distant search. *Academy of*
19
20 *Management Review*, 37(3), 355–375.
21
22
23
24 Agerström, J., & Rooth, D.-O. (2011). The role of automatic obesity stereotypes in real hiring
25
26 discrimination. *Journal of Applied Psychology*, 96(4), 790.
27
28
29
30 Agrawal, A., Cockburn, I., & Rosell, C. (2010). Not invented here? Innovation in company
31
32 towns. *Journal of Urban Economics*, 67(1), 78–89.
33
34
35
36 Ajzen, I. (2001). Nature and operation of attitudes. *Annual Review of Psychology*, 52(1), 27–58.
37
38
39 Albach, H., Pay, D., & Rojas, R. (1991). Quellen, Zeiten und Kosten von Innovationen. Deutsche
40
41 Unternehmen im Vergleich zu ihren japanischen und amerikanischen Konkurrenten. *Zeitschrift*
42
43 *für Betriebswirtschaft*, 61(3), 309–324.
44
45
46
47 Allen, T. J. (1977). Managing the flow of technology: Technology transfer and the dissemination
48
49 of technological information within the R&D organization. Cambridge, Mass: MIT Press.
50
51
52
53 Allen, T., Katz, R., Grady, J. ., & Slavin, N. (1988). Project team aging and performance: The
54
55 roles of project and functional managers. *R&D Management*, 18(4), 295–308.
56
57
58
59
60

- 1
2
3 Argote, L., McEvily, B., & Reagans, R. (2003). Managing Knowledge in Organizations: An
4 Integrative Framework and Review of Emerging Themes. *Management Science*, 49(4), 571–582.
5
6
7
8
9 Ashforth, B. E., & Mael, F. (1989). Social identity theory and the organization. *Academy of*
10
11 *Management Review*, 14(1), 20–39.
12
13
14 Baer, M., & Brown, G. (2012). Blind in one eye: How psychological ownership of ideas affects
15 the types of suggestions people adopt. *Organizational Behavior and Human Decision*
16 *Processes*, 118(1), 60–71.
17
18
19
20
21
22 Benner, M. J., & Tushman, M. L. (2003). Exploitation, Exploration, and Process Management:
23 The Productivity Dilemma Revisited. *Academy of Management Review*, 28(2), 238–256.
24
25
26
27
28 Biehl, M., Kim, H., & Wade, M. (2006). Relationships among the academic business disciplines:
29 a multi-method citation analysis. *Omega*, 34(4), 359–371.
30
31
32
33
34 Bitner, M. J., & Brown, S. W. (2008). The evolution and discovery of services science in
35 business schools. In *Services Science* (pp. 91–101). Springer.
36
37
38
39 Biyalogorsky, E., Boulding, W., & Staelin, R. (2006). Stuck in the Past: Why Managers Persist
40 with New Product Failures. *Journal of Marketing*, 70(2), 108–121.
41
42
43
44 Bohner, G., & Dickel, N. (2011). Attitudes and attitude change. *Annual Review of Psychology*,
45 62, 391–417.
46
47
48
49
50 Bohner, G., & Wänke, M. (2002). Attitudes and attitude change. Hove [etc.]: Psychology Press.
51
52
53
54 Bryman, A. (2008). *Social Research Methods*. Oxford; New York: Oxford University Press.
55
56
57
58
59
60

Bstieler, L., & Hemmert, M. (2010). Increasing Learning and Time Efficiency in Interorganizational New Product Development Teams. *Journal of Product Innovation Management*, 27(4), 485–499.

Burcharth, A. L. de A., Knudsen, M. P., & Søndergaard, H. A. (2014). Neither invented nor shared here: The impact and management of attitudes for the adoption of open innovation practices. *Technovation*, 34, 149–161.

Burrows, P. (2000). Apple: Yes, Steve, you fixed it! Congrats! Now what’s act two? *Business Week*, 3692, 102–112.

Cassiman, B., & Veugelers, R. (2006). In Search of Complementarity in Innovation Strategy: Internal R&D and External Knowledge Acquisition. *Management Science*, 52(1), 68–82.

Chang, Y.-Y., Gong, Y., & Peng, M. W. (2012). Expatriate Knowledge Transfer, Subsidiary Absorptive Capacity, and Subsidiary Performance. *Academy of Management Journal*, 55(4), 927–948.

Chen, S., & Chaiken, S. (1999). The heuristic-systematic model in its broader context. *Dual-Process Theories in Social Psychology*, 73–96.

Choliz, M. (2010). Cognitive Biases and Decision Making in Gambling. *Psychological Reports*, 107(1), 15–24.

Clagett, R. P. (1967). Receptivity to Innovation - Overcoming NIH. Massachusetts Institute of Technology, Massachusetts.

Cohen, W. M., & Levinthal, D. A. (1990). Absorptive Capacity: A New Perspective on Learning and Innovation. *Administrative Science Quarterly*, 35(1), 128–152.

- Crano, W. D., & Prislin, R. (2006). Attitudes and Persuasion. *Annual Review of Psychology*, 57(1), 345–374.
- Cucina, J. M., Martin, N. R., Vasilopoulos, N. L., & Thibodeaux, H. F. (2012). Self-Serving Bias Effects on Job Analysis Ratings. *Journal of Psychology*, 146(5), 511–531.
- Dahlander, L., & Gann, D. M. (2010). How open is innovation? *Research Policy*, 39(6), 699–709.
- Demski, R. M., & McGlynn, R. P. (1999). Fear or Moral Indignation? Predicting Attitudes Toward Parolees¹. *Journal of Applied Social Psychology*, 29(10), 2024–2058.
- Eagly, A. H., & Chaiken, S. (1993). The psychology of attitudes. Belmont, Calif: Wadsworth, Thompson.
- Eisenhardt, K. M. (1989). Agency theory: An assessment and review. *Academy of Management Review*, 14(1), 57–74.
- Elkins, T. J., Phillips, J. S., & Konopaske, R. (2002). Gender-related biases in evaluations of sex discrimination allegations: Is perceived threat the key? *Journal of Applied Psychology*, 87(2), 280–292.
- Elmquist, M., Fredberg, T., & Ollila, S. (2009). Exploring the field of open innovation. *European Journal of Innovation Management*, 12(3), 326–345.
- Englich, B., Mussweiler, T., & Strack, F. (2006). Playing dice with criminal sentences: The influence of irrelevant anchors on experts' judicial decision making. *Personality and Social Psychology Bulletin*, 32(2), 188–200.

Fazio, R. H. (1990). Multiple Processes by which Attitudes Guide Behavior: The Mode Model as an Integrative Framework. *Advances in Experimental Social Psychology*, 23, 75–109.

Fazio, R. H., & Olson, M. A. (2003). Implicit Measures in Social Cognition Research: Their Meaning and Use. *Annual Review of Psychology*, 54(1), 297–327.

Fey, C. F., & Furu, P. (2008). Top management incentive compensation and knowledge sharing in multinational corporations. *Strategic Management Journal*, 29(12), 1301–1323.

Franke, N., Keinz, P., & Steger, C. J. (2009). Testing the value of customization: when do customers really prefer products tailored to their preferences? *Journal of Marketing*, 73(5), 103–121.

Galinsky, A. D., & Moskowitz, G. B. (2000). Perspective-taking: decreasing stereotype expression, stereotype accessibility, and in-group favoritism. *Journal of Personality and Social Psychology*, 78(4), 708.

Gesing, J., Antons, D., Piening, E. P., Rese, M., & Salge, T. O. (2014). Joining Forces or Going it Alone? On the Interplay between External Collaboration Partner Types, Inter-firm Governance Modes and Internal R&D. *Journal of Product Innovation Management*.

Gigerenzer, G., & Gaissmaier, W. (2011). Heuristic decision making. *Annual Review of Psychology*, 62, 451–482.

Goldstein, D. G., & Gigerenzer, G. (2002). Models of Ecological Rationality. *Psychological Review*, 109(1), 75–90.

Green, S. E. (2004). A rhetorical theory of diffusion. *Academy of Management Review*, 29(4), 653–669.

- 1
2
3 Grimpe, C., & Kaiser, U. (2010). Balancing Internal and External Knowledge Acquisition: The
4
5 Gains and Pains from R&D Outsourcing. *Journal of Management Studies*, 47(8), 1483–1509.
6
7
8
9 Gupta, A. K., & Govindarajan, V. (2000). Knowledge flows within multinational corporations.
10
11 *Strategic Management Journal*, 21(4), 473–496.
12
13
14 Haas, M. R. (2010). The Double-Edged Swords of Autonomy and External Knowledge:
15
16 Analyzing Team Effectiveness in a Multinational Organization. *Academy of Management*
17
18 *Journal*, 53(5), 989–1008.
19
20
21
22 Harbring, C., & Irlenbusch, B. (2011). Sabotage in tournaments: Evidence from a laboratory
23
24 experiment. *Management Science*, 57(4), 611–627.
25
26
27
28 Hauser, J. (1998). Research, Development, and Engineering Metrics. *Management Science*,
29
30 44(12), 1670–1689.
31
32
33 Hauser, J., Tellis, G. J., & Griffin, A. (2006). Research on innovation: A review and agenda for
34
35 marketing science. *Marketing Science*, 25(6), 687–717.
36
37
38
39 Hayes, R. H., & Clark, K. B. (1985). Exploring the Source of Productivity Differences at the
40
41 Factory Level.
42
43
44 Hergovich, A., Schott, R., & Burger, C. (2010). Biased evaluation of abstracts depending on topic
45
46 and conclusion: Further evidence of a confirmation bias within scientific psychology. *Current*
47
48 *Psychology*, 29(3), 188–209.
49
50
51
52 Herzog, P., & Leker, J. (2010). Open and closed innovation - different innovation cultures for
53
54 different strategies. *International Journal of Technology Management*, 52(3/4), 322–343.
55
56
57
58
59
60

Hess, A. M., & Rothaermel, F. T. (2011). When are assets complementary? Star scientists, strategic alliances, and innovation in the pharmaceutical industry. *Strategic Management Journal*, 32(8), 895–909.

Hewstone, M., Rubin, M., & Willis, H. (2002). Intergroup Bias. *Annual Review of Psychology*, 53, 575–604.

Hölzl, E., Kirchler, E., & Rodler, C. (2002). Hindsight Bias in Economics Expectations: I Knew All Along What I Want to Hear. *Journal of Applied Psychology*, 87(3), 437–443.

Hunter, E. M., Perry, S. J., & Currall, S. C. (2011). Inside multi-disciplinary science and engineering research centers: The impact of organizational climate on invention disclosures and patents. *Research Policy*, 40(9), 1226–1239.

Hussinger, K., & Wastyn, A. (2011). In Search for the Not-Invented-Here Syndrome: The Role of Knowledge Sources and Firm Success. ZEW Discussion Paper No.11-048.

Husted, K., & Michailova, S. (2002). Diagnosing and Fighting Knowledge-Sharing Hostility. *Organizational Dynamics*, 31(1).

Im, S., Montoya, M. M., & Workman, J. P. (2013). Antecedents and Consequences of Creativity in Product Innovation Teams. *Journal of Product Innovation Management*, 30(1), 170–185.

Jacobs, J. A., & Frickel, S. (2009). Interdisciplinarity: A critical assessment. *Annual Review of Sociology*, 35, 43–65.

Jain, R. K., & Triandis, H. C. (1990). Management of research and development organizations. New York: Wiley.

- Jarell, S. B., & Stanley, T. D. (2004). Declining Bias and Gender Wage Discrimination? *The Journal of Human Resources*, 39(3), 828–838.
- Jeppesen, L. B., & Lakhani, K. R. (2010). Marginality and problem-solving effectiveness in broadcast search. *Organization Science*, 21(5), 1016–1033.
- Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2), 263–291.
- Kathoefer, D. (2012). Das Not-Invented-Here Syndrom in der Wissenschaft (1st ed.). Göttingen: Sierke-Verl.
- Kathoefer, D., & Leker, J. (2012). Knowledge transfer in academia: An exploratory study on the not-invented-here syndrome. *The Journal of Technology Transfer*, 37(5), 658–675.
- Katz, R., & Allen, T. J. (1982). Investigating the Not Invented Here (NIH) syndrome: A look at the performance, tenure, and communication patterns of 50 R&D Project Groups. *R&D Management*, 12(1), 7–20.
- Kelley, H. H. (1967). Attribution theory in social psychology. In D. Levine (Ed.), Nebraska symposium on motivation (pp. 192–238). Lincoln: University of Nebraska Press.
- Keysar, B., Hayakawa, S. L., & An, S. G. (2012). The Foreign-Language Effect Thinking in a Foreign Tongue Reduces Decision Biases. *Psychological Science*, 23(6), 661–668.
- Klein, K. J., & Sorra, J. S. (1996). The Challenge of Innovation Implementation. *Academy of Management Review*, 21(4), 1055–1080.

Koch, R., & Leitner, K.-H. (2008). The Dynamics and Functions of Self-Organization in the Fuzzy Front End: Empirical Evidence from the Austrian Semiconductor Industry. *Creativity & Innovation Management*, 17(3), 216–226.

Kostova, T. (1999). Transnational Transfer of Strategic Organizational Practices: A Contextual Perspective. *Academy of Management Review*, 24(2), 308–324.

Kostova, T., & Roth, K. (2002). Adoption of an Organizational Practice by Subsidiaries of Multinational Corporations: Institutional and Relational Effects. *Academy of Management Journal*, 45(1), 215–233.

Kozlowski, S. W. J., Chao, G. T., Grand, J. A., Braun, M. T., & Kuljanin, G. (2013). Advancing Multilevel Research Design Capturing the Dynamics of Emergence. *Organizational Research Methods*, 16(4), 581–615.

Laden, K. (1996). “Not Invented Here”, or, The Other Person’s Dessert Always Looks Better. *Research Technology Management*, 39(6), 10–12.

Lakatos, I. (1978). Falsification and the methodology of scientific research programmes. In I. Lakatos (Ed.), *The methodology of scientific research programmes* (pp. 8–101). Cambridge University Press Cambridge.

Lane, P. J., Koka, B. R., & Pathak, S. (2006). The Reification of Absorptive Capacity: A Critical Review and Rejuvenation of the Construct. *Academy of Management Review*, 31(4), 833–863.

Laursen, K., & Salter, A. (2006). Open for innovation: the role of openness in explaining innovation performance among U.K. manufacturing firms. *Strategic Management Journal*, 27(2), 131–150.

Leonard-Barton, D. (1992). Core capabilities and core rigidities: A paradox in managing new product development. *Strategic Management Journal*, 13(S1), 111–125.

Lichtenthaler, U. (2011). Open Innovation: Past Research, Current Debates, and Future Directions. *Academy of Management Perspectives*, 25(1), 75–93.

Lichtenthaler, U., & Ernst, H. (2006). Attitudes to externally organising knowledge management tasks: a review, reconsideration and extension of the NIH syndrome. *R&D Management*, 36(4), 367–386.

Lidwell, W., Holden, K., & Butler, J. (2010). Universal Principles of Design: 125 Ways to Enhance Usability, Influence Perception, Increase Appeal, Make Better Design Decisions, and Tech Through Design. Rockport Publishers.

Lucas, H. C. J., & Goh, J. M. (2009). Disruptive technology: How Kodak missed the digital photography revolution. *The Journal of Strategic Information Systems*, 18(1), 46–55.

Lüttgens, D., Antons, D., Pollok, P., & Piller, F. T. (2014). Wisdom of the Crowd and Capabilities of a Few: Internal Success Factors of Crowdsourcing for Innovation. *Journal of Business and Economics*, 84(3), 339–374.

Lyness, K. S., & Heilman, M. E. (2006). When Fit Is Fundamental: Performance Evaluations and Promotions of Upper-Level Female and Male Managers. *Journal of Applied Psychology*, 91(4), 777–785.

March, J. G. (1991). Exploration and exploitation in organizational learning. *Organization Science*, 2(1), 71–87.

McEvily, B., Perrone, V., & Zaheer, A. (2003). Trust as an Organizing Principle. *Organization Science*, 14(1), 91–103.

McKinlay, A., & Starkey, K. (1992). Strategy and human resource management. *International Journal of Human Resource Management*, 3(3), 435–450.

Mehrwald, H. (1999). Das “Not-Invented-Here”-Syndrom in Forschung und Entwicklung. Springer.

Menon, T., & Pfeffer, J. (2003). Valuing internal vs. external knowledge: Explaining the preference for outsiders. *Management Science*, 49(4), 497–513.

Menon, T., Thompson, L., & Choi, H.-S. (2006). Tainted Knowledge vs. Tempting Knowledge: People Avoid Knowledge from Internal Rivals and Seek Knowledge from External Rivals. *Management Science*, 52(8), 1129–1144.

Michailova, S., & Husted, K. (2003). Knowledge-sharing hostility in Russian firms. *California Management Review*, 45(3), 59–77.

Milkman, K. L., Chugh, D., & Bazerman, M. H. (2009). How Can Decision Making Be Improved? *Perspectives on Psychological Science*, 4(4), 379–383.

Morrison, E. (1966). *Men, Machines and Modern Times*. Cambridge: MIT Press.

Munro, G. D., & Ditto, P. H. (1997). Biased assimilation, attitude polarization, and affect in reactions to stereotype-relevant scientific information. *Personality and Social Psychology Bulletin*, 23(6), 636–653.

- Nelson, J. E., & Beggan, J. K. (2004). Self-Serving Judgments About Winning the Lottery. *Journal of Psychology*, 138(3), 253–264.
- Nickerson, R. S. (1998). Confirmation bias: a ubiquitous phenomenon in many guises. *Review of General Psychology*, 2(2), 175.
- Nosek, B. A., Hawkins, C. B., & Frazier, R. S. (2011). Implicit social cognition: from measures to mechanisms. *Trends in Cognitive Sciences*, 15(4), 152–159.
- Onarheim, B., & Christensen, B. T. (2012). Distributed idea screening in stage–gate development processes. *Journal of Engineering Design*, 23(9), 660–673.
- Pay, D. (1989). Kulturspezifische Determinanten der Organisation von Innovationsprozessen. Ein empirischer Vergleich in deutschen und amerikanischen Unternehmen. *Zeitschrift für Betriebswirtschaft (Ergänzungsheft)*, 1, 131–175.
- Pay, D. (1995). Organisationsmaßnahmen zur Verkürzung der Innovationszeit europäischer Unternehmen. *Zeitschrift für Betriebswirtschaft (Ergänzungsheft)*, (1), 77–102.
- Petty, R. E., & Wegener, D. T. (1998). Matching versus mismatching attitude functions: Implications for scrutiny of persuasive messages. *Personality and Social Psychology Bulletin*, 24(3), 227–240.
- Pierce, J. L., Kostova, T., & Dirks, K. T. (2001). Toward a Theory of Psychological Ownership in Organizations. *Academy of Management Review*, 26(2), 298–310.
- Pierce, S. J. (1991). Subject areas, disciplines, and the concept of authority. *Library & Information Science Research*, 13(1), 21–35.

1
2
3 Plott, C. R., & Zeiler, K. (2005). The Willingness to Pay-Willingness to Accept Gap, the
4 “Endowment Effect,” Subject Misconceptions, and Experimental Procedures for Eliciting
5 Valuations. *The American Economic Review*, 95(3), 530–545.
6
7
8
9
10
11 Pratkanis, A. R., & Greenwald, A. G. (1989). A sociocognitive model of attitude structure and
12 function. *Advances in Experimental Social Psychology*, 22, 245–285.
13
14
15
16
17 Read, D., & Grushka-Cockayne, Y. (2011). The similarity heuristic. *Journal of Behavioral*
18 *Decision Making*, 24(1), 23–46.
19
20
21
22 Reagans, R., & McEvily, B. (2003). Network structure and knowledge transfer: The effects of
23 cohesion and range. *Administrative Science Quarterly*, 48(2), 240–267.
24
25
26
27
28 Ritov, I., & Baron, J. (1990). Reluctance to vaccinate: Omission bias and ambiguity. *Journal of*
29 *Behavioral Decision Making*, 3(4), 263–277.
30
31
32
33 Rogan, M., & Mors, M. L. (2014). A Network Perspective on Individual-Level Ambidexterity in
34 Organizations. *Organization Science*.
35
36
37
38
39 Rosenberg, M. J., & Hovland, C. I. (1960). Cognitive, affective, and behavioral components of
40 attitudes. In C. I. Hovland & M. J. Rosenberg (Eds.), *Attitude organization and change: An*
41 *analysis of consistency among attitude components* (pp. 1–14). New Haven, CT: Yale University
42 Press.
43
44
45
46
47
48
49 Sadler-Smith, E., & Shefy, E. (2004). The intuitive executive: understanding and applying “gut
50 feel” in decision-making. *The Academy of Management Executive*, 18(4), 76–91.
51
52
53
54
55
56
57
58
59
60

1
2
3 Salomo, S., Keinschmidt, E. J., & De Brentani, U. (2010). Managing new product development
4 teams in a globally dispersed NPD program. *Journal of Product Innovation Management*,
5 27(7), 955–971.
6
7

8
9
10
11 Scott, S. G. (1997). Social identification effects in product and process development teams.
12 *Journal of Engineering and Technology Management*, 14(2), 97–127.
13
14

15
16
17 Shannon, C. E., & Weaver, W. (1964). The Mathematical Theory of Communication (11th ed.).
18 Urbana: The University of Illinois Press.
19

20
21
22 Sieg, J. H., Wallin, M. W., & von Krogh, G. (2010). Managerial challenges in open innovation: a
23 study of innovation intermediation in the chemical industry. *R&D Management*, 40(3), 281–291.
24
25

26
27
28 Simon, H. A. (1955). A behavioral model of rational choice. *The Quarterly Journal of*
29 *Economics*, 69(1), 99–118.
30
31

32
33
34 Skilton, P. F., & Dooley, K. J. (2010). The effects of repeat collaboration on creative abrasion.
35 *Academy of Management Review*, 35(1), 118–134.
36
37

38
39 Song, J., Asakawa, K., & Chu, Y. (2011). What determines knowledge sourcing from host
40 locations of overseas R&D operations?: A study of global R&D activities of Japanese
41 multinationals. *Research Policy*, 40(3), 380–390.
42
43
44

45
46
47 Song, M., & Swink, M. (2009). Marketing–manufacturing integration across stages of new
48 product development: Effects on the success of high-and low-innovativeness products. *IEEE*
49 *Transactions on Engineering Management*, 56(1), 31–44.
50
51
52

53
54
55 Staw, B. M. (1981). The escalation of commitment to a course of action. *Academy of*
56 *Management Review*, 6(4), 577–587.
57
58
59
60

Strack, F., Werth, L., & Deutsch, R. (2006). Reflective and impulsive determinants of consumer behavior. *Journal of Consumer Psychology*, 16(3), 205–216.

Szulanski, G. (1996). Exploring internal stickiness: Impediments to the transfer of best practice within the firm. *Strategic Management Journal*, 17(Sp. Iss. SI), 27–43.

Tajfel, H., & Turner, J. (1979). An Integrative Theory of Intergroup Conflict. In W. G. Austin & S. Worchel (Eds.), *Psychology of Intergroup Relations* (pp. 33–48). Monterey: Brooks-Cole.

Talke, K., Salomo, S., & Rost, K. (2010). How top management team diversity affects innovativeness and performance via the strategic choice to focus on innovation fields. *Research Policy*, 39(7), 907–918.

Toosi, N. R., Ambady, N., Babbitt, L. G., & Sommers, S. R. (2012). Dyadic interracial interactions: A meta-analysis. *Psychological Bulletin*, 138(1), 1–27.

Uhlmann, E. L., Leavitt, K., Menges, J. I., Koopman, J., Howe, M., & Johnson, R. E. (2012). Getting Explicit About the Implicit: A Taxonomy of Implicit Measures and Guide for Their Use in Organizational Research. *Organizational Research Methods*, 15(4), 553–601.

Von Hippel, E. (1988). *The Sources of Innovation*. New York, NY: Oxford Univ. Press.

Von Hippel, E. (2005). *Democratizing Innovation*. MIT Press.

Wood, G., Wilkinson, A., & Harcourt, M. (2008). Age discrimination and working life: perspectives and contestations—a review of the contemporary literature. *International Journal of Management Reviews*, 10(4), 425–442.

1
2
3 Wood, W. (2000). Attitude change: Persuasion and social influence. *Annual Review of*
4
5 *Psychology*, 51(1), 539–570.
6

7
8
9 Wuyts, S., Colombo, M. G., Dutta, S., & Nooteboom, B. (2005). Empirical tests of optimal
10
11 cognitive distance. *Journal of Economic Behavior & Organization*, 58, 277–302.
12

13
14 Zahra, S. A., & George, G. (2002). Absorptive Capacity: A Review, Reconceptualization, and
15
16 Extension. *Academy of Management Review*, 27(2), 185–203.
17

18
19
20 Zak, E. (2013, October 3). How Twitter's Hashtag Came to Be. WSJ Blogs - Digits.
21

22
23 Zardkoohi, A. (2004). Do real options lead to escalation of commitment? *Academy of*
24
25 *Management Review*, 29(1), 111–119.
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60

Appendix

An overview of focused research investigating "not invented here" (NIH)

Study	Research design	Sample	Definition of NIH	Operationalization	Examined boundary	Central results	Need for further research
Clagett, 1967	qualitative	8 case studies	"...NIH has been used among technical organizations to describe the attitude often spoke of as if it were a disease of technical organizations who resist adoption of an innovation proposed from a source outside of the organization." p. II	---	Organizational boundary, inter-functional and inter-disciplinary knowledge transfer	First evidence for existence of NIH	Approaches to persuade technical experts

Study	Research design	Sample	Definition of NIH	Operationalization	Examined boundary	Central results	Need for further research
Katz & Allen, 1982	quantitative	N = 345 employees from R&D working in 50 different R&D teams in an US-company	"...NIH syndrome is defined as the tendency of a project group of stable composition to believe it possesses a monopoly of knowledge of its field, which leads it to reject new ideas from outsiders to the likely detriment of its performance." p. 7	Decrease in rate of communication between a project group and external actors	Organizational boundary, actors external to team from inside and outside own company	Team tenure results in a decline of communication and use of communication tools and thus project performance	Group effects, moderating effects of management impact
Allen et al., 1988	quantitative	N = 2000 individuals from 181 R&D teams	Referring to Katz & Allen, 1982	Project performance decline due to team tenure	Organizational boundary	Project and functional manage rare central persons that may reduce NIH	---

Study	Research design	Sample	Definition of NIH	Operationalization	Examined boundary	Central results	Need for further research
Albach et al., 1991; Pay, 1989, 1995	qualitative and quantitative research as well as simulation	N=300 German, US and Japanese firms	Skepticism towards external results; referring to Katz & Allen, 1982	Open questions in survey, simulation study uses parameters to model effect of NIH on development time	Organizational and spatial boundary	NIH has greater impact in US-based than in German firms; and less impact in Japan than in Western culture	---
Mehrwald, 1999	quantitative	N=51 R&D manager and N= 89 developer employed at 53 German firms	A negative, invalid, generalizing and rigid attitude of individuals and groups respectively towards external technology that leads to suboptimal usage of external technology (p. 50)	New scale development to measure NIH of managers and developers	Organizational boundary	Negative correlation between NIH and effort to integrate external knowledge; however employees show heterogeneous NIH tendencies	Better conceptualization and operationalization of NIH construct; Need for larger sample sizes; Applying theoretical and methodological knowledge from social psychology

Study	Research design	Sample	Definition of NIH	Operationalization	Examined boundary	Central results	Need for further research
Lichtenthaler & Ernst, 2006	concept- ional	---	“...a negative attitude to knowledge that originates from a source outside the own institution.” (p. 368), referring to Clagett 1967, Katz & Allen, 1982 and Mehrwald, 1999	---	---	---	Large-scale empirical studies focusing on the syndrome
Agrawal et al., 2010	patent analysis	72 US regions and 264.078 patents	Referring to Katz & Allen, 1982	Degree of patent self-citation	Organizational and spatial boundary	Large and geographically isolated firms cite own patents more frequently and those patents receive less total citations	---

Study	Research design	Sample	Definition of NIH	Operationalization	Examined boundary	Central results	Need for further research
Hussinger & Wastyn, 2011	quantitative	N = 905 German manufacturing companies	<i>“The not-invented-here NIH syndrome refers to internal resistance against external knowledge”</i> (p. 1), referring to Clagett, 1967; Katz & Allen, 1982, Lichtenthaler & Ernst, 2006	Internal resistance towards innovation and knowledge acquisition	Organizational boundary	Internal resistance towards knowledge acquisition and collaboration with competitors exists	Project and team level studies to investigate whether complementary or substitutive knowledge gets rejected
Herzog & Leker, 2010	quantitative	N = 46 business units of a German chemical company	Referring to Mehrwald, 1999	One of scales proposed by Mehrwald, 1999	Organizational boundary	Business units with an open innovation strategy show less NIH tendencies	---

Study	Research design	Sample	Definition of NIH	Operationalization	Examined boundary	Central results	Need for further research
Kathoefer & Leker, 2012	quantitative	N=66 Austrian and N=170 German professors in physics and engineering	Referring to Lichtenthaler & Ernst, 2006	4 factors using 10 items	Organizational boundary	Both applied as well as scientific researches show NIH tendencies; basic researchers show less effects; collaborative projects reduce likelihood of NIH	Focus on different attitudinal components and antecedents of individual attitude development
Kathoefer, 2012	quantitative	N=477 German professors in biology, physics and chemistry	<i>“The Not-Invented-Here-Syndrome refers to a systematic negative attitude of an individual towards external knowledge.”</i> (p. 76)	Operationalization using three scales, two of them according to Mehrwald, 1999	Contextual and organizational boundary	NIH exists in academia and is result of opinions of colleagues, group pride, frequency of external communication, quality and number of former projects	Transferring findings to more applied domains like engineering

Study	Research design	Sample	Definition of NIH	Operationalization	Examined boundary	Central results	Need for further research
Burcharth et al., 2014	quantitative	N=331 manufacturing firms from Denmark	Referring to Katz & Allen, 1982 and Lichtenthaler & Ernst, 2006	Third-person approach reporting average employee attitude towards external knowledge on a 4 item scale	Organizational boundary	NIH tendencies in the R&D team lead to less adoption of inbound open innovation activities, creativity training limits while nursing of special talents fosters NIH	Studies following a longitudinal approach and focusing on individual instead of team-level attitudes are needed

Biographies

David Antons (antons@time.rwth-aachen.de) is assistant professor at the Innovation, Strategy and Organisation Group at RWTH Aachen University, where he received his Ph.D. (with distinction) working in close cooperation with the Institute of Psychology. His research examines organizational, spatial, and disciplinary interfaces, individual decision making, learning from failure, and (implicit) attitude measurement.

Frank Piller (piller@time.rwth-aachen.de) is a professor of management and the head of the Technology and Innovation Management Group at RWTH Aachen University. His research focuses on managing innovation interfaces, innovation in open business models, and novel modes of technology transfer.